Surface Area of Prisms

Lesson 10-4

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Paint every outside surface of a box — the total painted area is the surface area

Surface area

The total area of all the flat sides of a solid.

A box with 6 flat rectangle sides: top, bottom, front, back, left, right

Rectangular prism

A solid box shape with six flat rectangle sides.

Shaped like a tent or a wedge of cheese — 2 triangle ends + 3 rectangle sides

Triangular prism

A solid with two triangle ends and three flat rectangle sides.

A triangular prism has 5 faces: 2 triangles + 3 rectangles

Face

One flat side of a solid shape.

Unfold a triangular prism flat: you see 2 triangles and 3 rectangles side by side

Net

A flat shape that folds up into a solid.

On a triangular prism, the 3 rectangles that wrap around the sides are lateral faces

Lateral face

The side faces of a solid, not the top or bottom.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the surface area of rectangular and triangular prisms.

1️⃣ Notice

Your class is painting the outside of different time capsule shapes to protect them from weather.

2️⃣ Key idea

I can find the surface area of rectangular and triangular prisms.

3️⃣ Words I'll use

Surface area
Rectangular prism
Triangular prism
Face
Net
Lateral face

🤔 Think About It

- How many faces does a rectangular prism have? A triangular prism?
- What shapes make up the faces of a triangular prism?
- How is finding surface area different from finding volume?

👀 Watch

Watch your teacher model one example. Jot what you see.

____

🤝 We try

Solve the next one together as a class.

____

____

✏️ You try

Now try one on your own.

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Your class is painting two capsule shapes: a rectangular prism and a triangular prism. How is finding the surface area of each shape the same, and how is it different?

Level 1 support

Start here: For both, you add the areas of all the faces, but the shapes and number of faces differ.

💡 **Need a hint?**

Sentence starters (optional)

Both surface areas are found by adding the areas of all the ____.

Ambas áreas de superficie se hallan sumando las áreas de todas las ____.

A rectangular prism has ____ faces, but a triangular prism has ____ faces.

Un prisma rectangular tiene ____ caras, pero un prisma triangular tiene ____ caras.

Word bank: surface area | rectangular prism | triangular prism | face | lateral face

Listen for: *Students explain both methods add up all face areas, but a rectangular prism has 6 rectangular faces while a triangular prism has 5 faces (2 triangles + 3 rectangles).*

Level 2

Why does surface area use square units while volume uses cubic units? Justify with what each measurement counts.

- Surface area uses square units because it measures ____.
- Volume uses cubic units because it measures ____.

EXPLORE

For the triangular prism (base 6, height 4, length 10, slant sides 5), how did you account for the triangular bases AND the rectangular lateral faces?

Level 1 support

Start here: Add the two triangular bases to the three rectangular lateral faces.

💡 **Need a hint?**

Sentence starters (optional)

The two triangular bases together are 2 × (½ × 6 × 4) = ____.

Las dos bases triangulares juntas son 2 × (½ × 6 × 4) = ____.

The three lateral faces add to ____, so the total surface area is ____.

Las tres caras laterales suman ____, así que el área total de superficie es ____.

Word bank: surface area · triangular prism · lateral face · face · net

Listen for: *Students compute the two triangle bases (24 in²) and the three rectangles (60 + 50 + 50 = 160 in²), totaling SA = 184 in², and distinguish bases from lateral faces.*

Level 2

Compare the rectangular capsule (SA = 248 in²) with the similar triangular capsule (SA = 184 in²). Why does the rectangular one need more paint?

- The rectangular prism needs more paint because it has ____ faces and ____.
- Even at a similar size, the triangular prism uses less because ____.

CONNECT

A carpenter stains a chest that is 3 ft × 2 ft × 2 ft, and one can covers 50 ft^2. Talk through how to find the surface area and how many cans are needed.

Level 1 support

Start here: Find the surface area with SA = 2lw + 2lh + 2wh, then divide by 50 and round up.

💡 **Need a hint?**

Sentence starters (optional)

The chest's surface area is ____ ft^2 because SA = 2(3×2) + 2(3×2) + 2(2×2).

El área de superficie del cofre es ____ ft^2 porque SA = 2(3×2) + 2(3×2) + 2(2×2).

The carpenter needs ____ can(s) because ____ ÷ 50 = ____.

El carpintero necesita ____ lata(s) porque ____ ÷ 50 = ____.

Word bank: surface area | rectangular prism | face | lateral face | net

Listen for: *Students compute SA = 12 + 12 + 8 = 32 ft^2, then 32 ÷ 50 < 1, and reason that 1 can is enough since the surface area is under 50 ft^2.*

Level 2

If the chest had a lid that does not get stained, how would that change the surface area you calculate? Explain.

- Leaving the lid unstained changes the surface area by ____ because ____.
- I would subtract ____ from the total because ____.

REFLECT

A classmate wrote SA = 2(12×7) + 2(12×5) + (7×5) = 323 cm² for a 12 × 7 × 5 prism. How do you find and fix their mistake?

Level 1 support

Start here: They forgot to multiply the last face pair by 2; the correct SA is 358 cm².

💡 **Need a hint?**

Sentence starters (optional)

The mistake is that the last pair (7×5) was not multiplied by ____.

El error es que el último par (7×5) no se multiplicó por ____.

The correct surface area is ____ cm² because 2(7×5) = ____, not ____.

El área de superficie correcta es ____ cm² porque 2(7×5) = ____, no ____.

Word bank: surface area | rectangular prism | face | lateral face | net

Listen for: *Students notice the missing '2' on the last pair, compute 2(7×5) = 70, and correct the total to 168 + 120 + 70 = 358 cm².*

Level 2

What general rule about prism faces does this error break, and how could the net help a student avoid it?

- The error breaks the rule that opposite faces ____.
- Drawing the net would help because ____.

Guided Examples

Example 1

What is the surface area of a rectangular prism with $l = 5$ cm, $w = 4$ cm, $h = 3$ cm?

Solution: $SA = 2(5 \times 4) + 2(5 \times 3) + 2(4 \times 3) = 40 + 30 + 24 = 94$ cm^2.

Answer: A. 94 cm^2

Example 2

How many faces does a triangular prism have?

Solution: A triangular prism has 5 faces: 2 triangular bases and 3 rectangular lateral faces.

Answer: A. 5

Example 3

A triangular prism has two triangular bases each with area 12 cm^2 and three rectangular faces with areas 30 cm^2, 40 cm^2, and 40 cm^2. What is the total surface area?

Solution: $SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134$ cm^2.

Answer: A. 134 cm^2

Write About the Math

The Writing Revolution

I can explain my work using the words surface area, rectangular prism, triangular prism, and face.

1. Kernel Sentence

subject + verb

Model: Surface area is the total area of all the flat sides of a solid.
Área de superficie es el área total de todos los lados planos de un sólido.

Write a kernel sentence about surface area. Use a subject and a verb.
Escribe una oración base sobre área de superficie. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Surface area matters in math
Área de superficie importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Surface area matters in math because ____.
Área de superficie importa en matemáticas porque ____.

____

but
pero

Surface area matters in math, but ____.
Área de superficie importa en matemáticas, pero ____.

____

so
entonces

Surface area matters in math, so ____.
Área de superficie importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about surface area.
Di un hecho verdadero sobre surface area.

Surface area ____.

Question
Pregunta

Ask a question about surface area.
Haz una pregunta sobre surface area.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about surface area.
Muestra entusiasmo sobre surface area.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with surface area.
Dile a un compañero qué hacer con surface area.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

Both surface areas are found by adding the areas of all the ____.
Ambas áreas de superficie se hallan sumando las áreas de todas las ____.

A rectangular prism has ____ faces, but a triangular prism has ____ faces.
Un prisma rectangular tiene ____ caras, pero un prisma triangular tiene ____ caras.

The two triangular bases together are $2 \times (\frac{1}{2} \times 6 \times 4) =$ ____.
Las dos bases triangulares juntas son $2 \times (\frac{1}{2} \times 6 \times 4) =$ ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A triangular prism has two triangular bases each with area 12 cm^2 and three rectangular faces with areas 30 cm^2, 40 cm^2, and 40 cm^2. What is the total surface area?

A. 134 cm^2

B. 110 cm^2

C. 122 cm^2

D. 134 cm^3

Show your work:

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2. A triangular prism has triangular bases with base 6 in and height 4 in. The prism is 10 in long and the two slant sides of the triangle are each 5 in. What is the surface area?

A. 184 in^2

B. 160 in^2

C. 124 in^2

D. 184 in^3

Show your work:

Stretch Your Thinking Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A company ships products in two container options: a rectangular prism (8 × 6 × 4 in) or a triangular prism (triangle base 8 in, triangle height 6 in, prism length 8 in, slant sides 6.3 in each). Both must be wrapped in protective film. Which shape uses LESS film? Show your work.

Sentence starter: Rectangular prism SA = 2(____) + 2(____) + 2(____) = ____ in². Triangular prism SA = 2(____) + ____ + ____ + ____ = ____ in². The ____ uses less film because ____.

Show your work:

____

____

____

____

Reflect — Exit Ticket

A rectangular prism has l = 8 ft, w = 6 ft, h = 3 ft. What is the surface area?

A. 180 ft^2
B. 144 ft^2
C. 180 ft^3
D. 90 ft^2

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **Try It 1:** A. 134 cm^2 — *SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134* cm^2.
2. **Try It 2:** A. 184 in^2 — *Triangular bases:* $2 \times (\frac{1}{2} \times 6 \times 4) = 24$ in^2. *Three rectangular faces:* $6(10) + 5(10) + 5(10) = 60 + 50 + 50 = 160$ in^2. *SA = 24 + 160 = 184* in^2.
3. **Exit Ticket:** A. 180 ft^2 — *SA =* $2(8\times6) + 2(8\times3) + 2(6\times3) = 96 + 48 + 36 = 180$ ft^2. *Surface area uses square units* (ft^2).

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Surface area is the total area of all the flat sides of a solid.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).